MATH3026/4022 Lecture 12
Forecasting II

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Outline

- Forecasting for the $MA(2)$ process
- Forecasting for the $ARMA(1,1)$ process
- Forecasting for the $ARIMA(1,1,1)$ process
- Forecasting using R

Introduction

In Lecture 11 we introduced the concept of forecasting/prediction using time series models.

We saw how minimisation of the mean square prediction error can be used to derive future predictions for ARMA models and considered some specific examples: $AR(1)$, $AR(2)$ and $AR(p)$ models.

In this lecture, we continue this work, looking at some further examples and learning how forecasting may be carried out using `R`.

Forecasting: $MA(2)$ process

First, we consider prediction for the $MA(2)$ process

$$X_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}.$$

We note that the above process can be written in $MA(\infty)$ form as

$$X_t = \sum_{j=0}^{\infty} b_j Z_{t-j}$$

where $b_0 = 1, b_1 = \theta_1, b_2 = \theta_2$ and $b_j = 0$ where $j > 2$.

Once again, we require the predicted/forecasted value of the series at time $n+k$ ($k > 0$, having observed the time series to time $t = n$), written in the form

$$\hat{X}_{n+k}(n) = \sum_{j=0}^{\infty} d_j(k) X_{n-j}.$$

Forecasting: $MA(2)$ process

Having obtained, the values of $b_1, \ldots, b_{k-1}$, we need to find the values of $d_j(k)$ to make the prediction

$$\hat{X}_{n+k}(n) = \sum_{j=0}^{\infty} d_j(k) X_{n-j}.$$

To do this, we define the following functions

$$B(\xi) = \sum_{m=0}^{\infty} b_m \xi^m$$

$$C_k(\xi) = \sum_{j=0}^{\infty} c_j(k) \xi^j = \sum_{j=0}^{\infty} b_{j+k} \xi^j$$

$$D_k(\xi) = \sum_{j=0}^{\infty} d_j(k) \xi^j.$$

Forecasting: $MA(2)$ process

Each of the functions above is a generating function and the b_j coefficients are known because these are coefficients from the $MA(\infty)$ representation of the model.

It can be shown that

$$C_k(\xi) = B(\xi) D_k(\xi).$$

Then

$$D_k(\xi) = \frac{C_k(\xi)}{B(\xi)} = \frac{\sum_{j=0}^{\infty} b_{j+k}\xi^j}{\sum_{m=0}^{\infty} b_m \xi^m}.$$

The above may be expanded out as a polynomial with terms $\xi, \xi^2, \ldots$ and then the coefficient of ξ^j in $D_k(\xi)$ is $d_j(k)$.

Forecasting: $MA(2)$ process

We define the generating function $B(\xi)$ where

$$\begin{aligned} B(\xi) &= \sum_{j=0}^{\infty} b_j \xi^j \\ &= 1 + \theta_1 \xi + \theta_2 \xi^2 \\ &= (1 - \delta_1 \xi)(1 - \delta_2 \xi) \ \ \text{(assuming unequal roots).} \end{aligned}$$

then

$$\begin{aligned} B(\xi)^{-1} &= \left[\frac{1}{1 - \delta_1 \xi}\right]\left[\frac{1}{1 - \delta_2 \xi}\right] \\ &= \sum_{j=0}^{\infty} \left(\frac{\delta_1^{j+1} - \delta_2^{j+1}}{\delta_1 - \delta_2}\right) \xi^j. \end{aligned}$$

Forecasting: $MA(2)$ process

We seek

$$D_k(\xi) = \sum_{j=0}^{\infty} d_j(k)\xi^j = \frac{\sum_{j=0}^{\infty} b_{j+k}\xi^j}{B(\xi)} \quad (*)$$

to obtain the $d_j(k)$ terms, because

$$\hat{X}_{n+k}(n) = d_0(k)X_n + d_1(k)X_{n-1} + d_2(k)X_{n-2} + \dots$$

However, solution of (*) does not appear to be straightforward so we consider some particular cases...

Forecasting: $MA(2)$ process

Suppose that $k > 2$, then in (*)

$$\sum_{j=0}^{\infty} b_{j+k}\xi^j = b_k + b_{k+1}\xi + b_{k+2}\xi^2$$

$$= 0 \quad \text{(because } b_k = 0 \text{ for } k > 2\text{)}.$$

This implies that, when $k > 2$:

$$d_j(k) = 0, \quad j = 0, 1, 2, \dots$$

Therefore, the forecasted value of the series at time $n + k$ is

$$\hat{X}_{n+k}(n) = 0.$$

Note here that the forecasted value is just the mean of the process.

Forecasting: $MA(2)$ process

Suppose that $k = 2$, then in (*)

$$\begin{aligned}
D_2(\xi) &= \frac{\sum_{j=0}^{\infty} b_{j+2}\xi^j}{B(\xi)} = \frac{b_2 + b_3\xi + b_4\xi^2}{B(\xi)} \\
&= \frac{\theta_2}{B(\xi)} \\
&= \sum_{j=0}^{\infty} \left(\theta_2 \left(\frac{\delta_1^{j+1} - \delta_2^{j+1}}{\delta_1 - \delta_2} \right) \right) \xi^j
\end{aligned}$$

$d_j(2)$ is the coefficient of ξ^j in the above sum, hence

$$d_j(2) = \theta_2 \left(\frac{\delta_1^{j+1} - \delta_2^{j+1}}{\delta_1 - \delta_2} \right).$$

and $\hat{X}_{n+2}(n) = \sum_{j=0}^{\infty} d_j(2) X_{n-j}$.

Suppose that $k = 1$, then in (*)

$$
\begin{aligned}
D_1(\xi) &= \frac{\sum_{j=0}^{\infty} b_{j+1}\xi^j}{B(\xi)} = \frac{b_1 + b_2\xi + b_3\xi^2}{B(\xi)} \\
&= \frac{\theta_1 + \theta_2\xi}{B(\xi)} \\
&= (\theta_1 + \theta_2\xi)\left[\sum_{j=0}^{\infty}\left(\frac{\delta_1^{j+1} - \delta_2^{j+1}}{\delta_1 - \delta_2}\right)\xi^j\right] \\
&= \frac{1}{\delta_1 - \delta_2}\sum_{j=0}^{\infty}\theta_1(\delta_1^{j+1} - \delta_2^{j+1})\xi^j + \theta_2(\delta_1^{j+1} - \delta_2^{j+1})\xi^{j+1} \\
&= \frac{1}{\delta_1 - \delta_2}\left\{\theta_1(\delta_1 - \delta_2) + \sum_{u=1}^{\infty}\left[\theta_1(\delta_1^{u+1} - \delta_2^{u+1}) + \theta_2(\delta_1^{u} - \delta_2^{u})\right]\xi^u\right\}
\end{aligned}
$$

Forecasting: $MA(2)$ process

Examining the coefficients of ξ^u, we see that

$$d_u(1) = \begin{cases} \theta_1 & u = 0; \\ \frac{1}{\delta_1 - \delta_2}\left[\theta_1(\delta_1^{u+1} - \delta_2^{u+1}) + \theta_2(\delta_1^u - \delta_2^u)\right] & u \geq 1. \end{cases}$$

Using the above coefficients

$$\hat{X}_{n+1}(n) = \sum_{u=0}^{\infty} d_u(1) X_{n-u}.$$

Example 12.1

Consider the $MA(2)$ process

$$X_t = Z_t - \frac{7}{12}Z_{t-1} + \frac{1}{12}Z_{t-2}.$$

Suppose that the process has been observed up to time $t = n$. Derive the form of the forecasted value at time $t = n + k$ (assuming that the forecasted value is that which minimises the mean square prediction error) for

(a) k=1;

(b) k=2.

Example 12.1

Example 12.1

Example 12.1

Forecasting: $ARMA(1,1)$ process

Recall that the $ARMA(1,1)$ process is written

$$(1-\phi B)X_t = (1+\theta B)Z_t$$

which implies that

$$\begin{aligned} X_t &= \phi X_{t-1} + Z_t + \theta Z_{t-1} \\ &= \sum_{j=0}^{\infty} b_j Z_{t-j} \end{aligned}$$

since X_{t-1} can be written in $MA(\infty)$ form. Note here that the b_j terms are functions of θ and ϕ.

Forecasting: $ARMA(1,1)$ process

As we saw in Lecture 11

$$\begin{aligned} X_{n+k} &= \sum_{j=0}^{\infty} b_j Z_{n+k-j} \\ &= \underbrace{\sum_{j=0}^{k-1} b_j Z_{n+k-j}}_{\text{Future error}} + \underbrace{\sum_{j=k}^{\infty} b_j Z_{n+k-j}}_{\text{Process up to time } n} \\ &= \text{Future error} + \hat{X}_{n+k}(n). \end{aligned}$$

Forecasting: $ARMA(1,1)$ process

We may deduce

$$\hat{X}_{n+k}(n) = \sum_{j=0}^{\infty} b_j [Z_{n+k-j}]$$

with

$$[Z_{n+k}] = \begin{cases} 0 & k > 0 \\ Z_{n+k} & k \leq 0. \end{cases}.$$

and, for an $ARMA(1,1)$ process, we know that

$$X_{n+k} - \phi X_{n+k-1} = Z_{n+k} + \theta Z_{n+k-1}.$$

It follows that

$$\hat{X}_{n+k} - \phi \hat{X}_{n+k-1} = [Z_{n+k}] + \theta [Z_{n+k-1}].$$

Forecasting: $ARMA(1,1)$ process

Where $k = 1$

$$\begin{aligned}
\hat{X}_{n+1}(n) &= \phi \hat{X}_n(n) + [Z_{n+1}] + \theta [Z_n] \\
&= \phi X_n + 0 + \theta Z_n \quad \text{(because } \hat{X}_n(n) = X_n\text{)} \\
&= \phi X_n + \theta Z_n.
\end{aligned}$$

Where $k = 2$

$$\begin{aligned}
\hat{X}_{n+2}(n) &= \phi \hat{X}_{n+1}(n) + [Z_{n+2}] + \theta [Z_{n+1}] \\
&= \phi \hat{X}_{n+1}(n) \\
&= \phi \left(\phi X_n + \theta Z_n \right) \\
&= \phi^2 X_n + \theta \phi Z_n.
\end{aligned}$$

Forecasting: $ARMA(1,1)$ process

Generally, for $k \geq 2$

$$\begin{aligned}
\hat{X}_{n+k}(n) &= \phi \hat{X}_{n+k-1}(n) + [Z_{n+k}] + \theta [Z_{n+k-1}] \\
&= \phi \hat{X}_{n+k-1}(n) \\
&= \phi^{k-1} \left(\phi X_n + \theta Z_n \right)
\end{aligned}$$

At this point, we require Z_n as a function of $X_n, X_{n-1}, \ldots$ to evaluate the above expression. Note that

$$\begin{aligned}
Z_n &= (1+\theta B)^{-1}(1-\phi B) X_n \\
&= (X_n - \phi X_{n-1}) \sum_{j=0}^{\infty} (-\theta B)^j \\
&= X_n - (\phi+\theta) X_{n-1} + \theta(\phi+\theta) X_{n-2} - \ldots
\end{aligned}$$

Forecasting: $ARMA(1,1)$ process - Derivation

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Forecasting: $ARMA(1,1)$ process

Then

$$\begin{aligned}
\hat{X}_{n+k}(n) &= \phi^{k-1}\left(\phi X_n + \theta(X_n - (\phi+\theta)X_{n-1} + \theta(\phi+\theta)X_{n-2} - \ldots)\right) \\
&= \phi^{k-1}(\phi+\theta)\left(X_n - \theta X_{n-1} + \theta^2 X_{n-2} - \theta^3 X_{n-3} + \ldots\right) \\
&= \phi^{k-1}(\phi+\theta)\sum_{j=0}^{\infty}(-\theta)^j X_{n-j}
\end{aligned}$$

Note that

$$\lim_{k\to\infty} \hat{X}_{n+k}(n) = 0.$$

We see that, in the long term, the predicted value tends towards the mean of the process.

Forecasting: $ARIMA(1,1,1)$ process

Now, we'll explore how to forecast when a process is non-stationary, using the $ARIMA(1,1,1)$ process as an example.

Recall that the $ARIMA(1,1,1)$ process is written

$$\begin{aligned} (1-\phi B)\nabla X_t &= (1+\theta B)Z_t \\ \implies (1-\phi B)(1-B)X_t &= (1+\theta B)Z_t \end{aligned}$$

Rather than developing new theory for the non-stationary series X_t, we note that $Y_t = \nabla X_t = (1-B)X_t$ is a stationary series.

Moreover, we know that Y_t is an $ARMA(1,1)$ process so we may apply what we've learned about forecasting for an $ARMA(1,1)$ to Y_t and then transform forecasted values of Y_t to obtain forecasted values of X_t.

Forecasting: $ARIMA(1,1,1)$ process

If $\hat{Y}_{n+k}(n)$ is the predicted value of Y_t at $t = n + k$ then

$$\begin{aligned}\hat{Y}_{n+k}(n) &= (1-B)\hat{X}_{n+k}(n) \\ &= \hat{X}_{n+k}(n) - \hat{X}_{n+k-1}(n)\end{aligned}$$

Rearranging, we see that

$$\hat{X}_{n+k}(n) = \hat{Y}_{n+k}(n) + \hat{X}_{n+k-1}(n).$$

Hence

$$\hat{X}_{n+1}(n) = \hat{Y}_{n+1}(n) + \hat{X}_{n}(n)$$

and, using the above

$$\begin{aligned}\hat{X}_{n+2}(n) &= \hat{Y}_{n+2}(n) + \hat{X}_{n+1}(n) \\ &= \hat{Y}_{n+2}(n) + \hat{Y}_{n+1}(n) + \hat{X}_{n}(n)\end{aligned}$$

Forecasting: $ARIMA(1,1,1)$ process

In general, we see that

$$\hat{X}_{n+k}(n) = \hat{Y}_{n+k}(n) + \hat{Y}_{n+k-1}(n) + \ldots + \hat{Y}_{n+1}(n) + \hat{X}_n(n)$$

Note that $\hat{Y}_{n+k}(n), \ldots, \hat{Y}_{n+1}(n)$ would be forecasted values from an $ARMA(1,1)$ process (covered earlier) and that $\hat{X}_n(n)$ would be replaced with X_n since this value of X_t is observed.

Next we consider how to compute the prediction error when forecasting from an $ARIMA(1,1,1)$ process.

Forecasting: $ARIMA(1,1,1)$ process

We define the prediction error at time $n+k$, having observed the series up to time n, as

$$e_{n+k} = X_{n+k} - \hat{X}_{n+k}(n).$$

Then,

$$\begin{aligned}(1-B)e_{n+k} &= (1-B)(X_{n+k} - \hat{X}_{n+k}(n)) \\ &= (1-B)X_{n+k} - (1-B)\hat{X}_{n+k}(n) \\ &= Y_{n+k} - \hat{Y}_{n+k}(n)\end{aligned}$$

Recall that

$$Y_{n+k} = \sum_{j=0}^{\infty} b_j Z_{n+k-j} \quad \text{and} \quad \hat{Y}_{n+k}(n) = \sum_{j=0}^{\infty} b_j [Z_{n+k-j}]$$

for some coefficients $\{b_j\}$.

Then

$$
\begin{aligned}
(1-B)\,e_{n+k} &= \sum_{j=0}^{\infty} b_j Z_{n+k-j} - \sum_{j=0}^{\infty} b_j [Z_{n+k-j}] \\
&= Z_{n+k} + b_1 Z_{n+k-1} + \ldots + b_{k-1} Z_{n+1} \\
&= f_{n+k}
\end{aligned}
$$

where

$$
f_{t+k} = \sum_{j=0}^{k-1} b_j Z_{t+k-j} \quad \text{(with } b_0 = 1\text{)}.
$$

Forecasting: $ARIMA(1, 1, 1)$ process

We see that

$$\begin{aligned} e_{n+k} - e_{n+k-1} &= f_{n+k} \\ \implies e_{n+k} &= e_{n+k-1} + f_{n+k} \end{aligned}$$

It follows that

$$\begin{aligned} e_{n+1} &= e_n + f_{n+1} \\ &= f_{n+1} \quad \text{(since } X_n \text{ is observed)} \end{aligned}$$

Then

$$\begin{aligned} e_{n+2} &= e_{n+1} + f_{n+2} \\ &= f_{n+1} + f_{n+2}. \end{aligned}$$

We deduce that

$$e_{n+k} = f_{n+k} + f_{n+k-1} + \ldots + f_{n+1}.$$

Forecasting: $ARIMA(1,1,1)$ process

Replacing f_j terms with the functions of Zs, we obtain

$$\begin{aligned} e_{n+k} = Z_{n+k} &+ (1+b_1)Z_{n+k-1} + (1+b_1+b_2)Z_{n+k-2} \\ &+ \ldots + (1+b_1+\ldots+b_{k-1})Z_{n+k}. \end{aligned}$$

Then

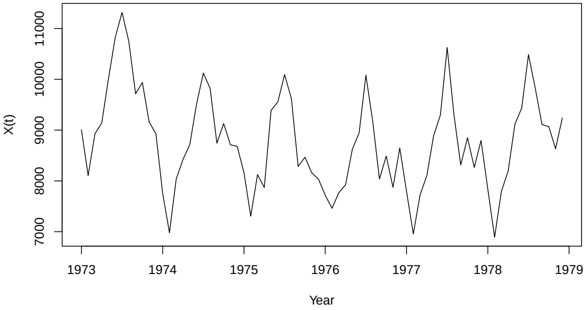
$$\begin{aligned} \sigma_k^2 &= \mathbb{E}(e_{n+k}^2) \\ &= \sigma_z^2\left[1+(1+b_1)^2+\ldots+(1+b_1+\ldots+b_{k-1})^2\right] \\ &= \sigma_{k-1}^2 + \sigma_z^2(1+b_1+\ldots+b_{k-1})^2 \end{aligned}$$

and we note that $\sigma_k^2 \to \infty$ as k gets very large, thereby implying that long-term forecasting for non-stationary time series is difficult and involves significant uncertainty.

Forecasting in R

The R package `forecast` can be used to make forecasts from time series models.

We recall the monthly US accidental deaths data from 1973–1978 and suppose that we'd like to forecast the series for six months for January–June 1979.



The data exhibit obvious seasonality and so we fit a

$$ARIMA(0,1,1) \times (0,1,1)_{12}$$

to the data.

The next slide shows the R code used to fit this model and to forecast values for January–June 1979.

R Code

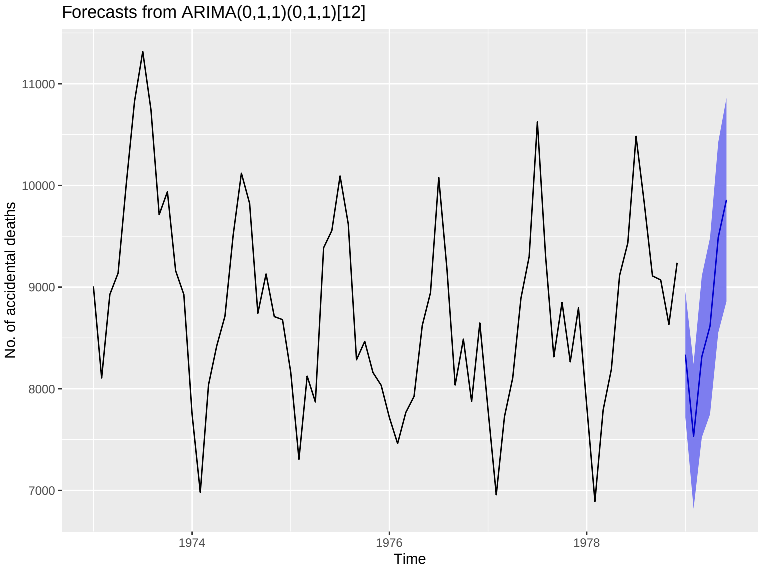
```r
#Read in the data
x<-USAccDeaths

#Fit the SARIMA model
mod1<-arima(x,order=c(0,1,1),seasonal=list(order=c(0,1,1),
             period=12))

#Forecast for the next 6 months (h=6)
fc_6m<-forecast(x,h=6,model=mod1,level=95)

#Plot the forecasted values
autoplot(fc_6m)
```

Forecasting in R



Forecasting in R

The true numbers of accidental deaths for January–June 1979 are known:

$$7798, 7406, 8363, 8460, 9217, 9316.$$

Now we'd like to see how our forecasted values compare to the observed values.

To do this, we produce a plot that shows both the true values and the forecasted values.

The R code to do this is shown on the next slide, followed by the corresponding plot.

Forecasting in R

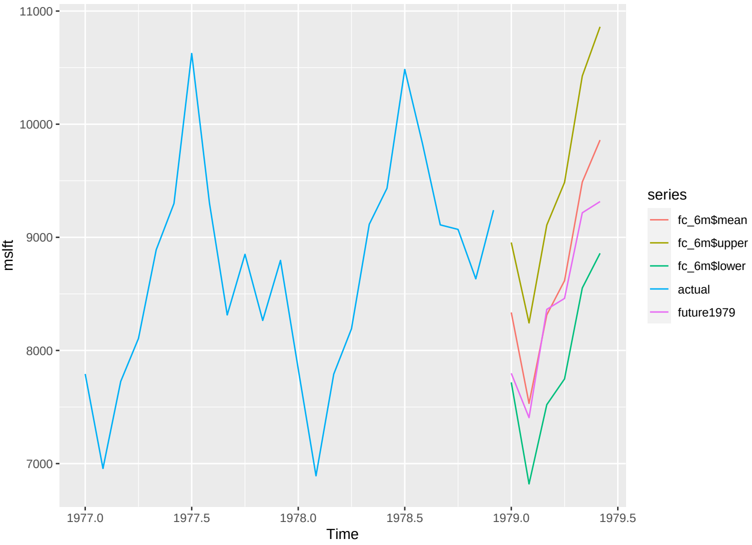
```
#We extract the values of the series for 1977 and 1978
deaths77_78<-ts(USAccDeaths[49:72],start=1977,frequency=12)

#We read in the true number of accidental deaths in 1979
#as a time series of length 6
future1979<-ts(c(7798,7406,8363,8460,9217,9316),frequency=12,
start=1979)

#Using the model and forecasts fitted earlier (slide 34)
#combine all series into one object, for plotting
USAcc_fit<-cbind(fc_6m$mean,fc_6m$upper,fc_6m$lower,
deaths77_78,future1979)

#Plot the true values for 1977-78 and the forecasted and true
#values for the first six months of 1979 (with conf. int.)
autoplot(USAcc_fit)
```

Forecasting in R



Lecture 12: Learning Outcomes

- Understand how predictions at time $n+k$ may be made for the following processes, having observed data $(x_1, \ldots, x_n)$:
  - $MA(2)$ process
  - $ARMA(1,1)$ process
  - $ARIMA(1,1,1)$ process
- Be able to forecast from models using R and interpret results.